# Week 5 Notes

Exponential and Logarithmic Functions:
One-to-One Functions, Line Tests, Exponential and Composite Functions

**Course:** Mathematics–1

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## Lecture 1 — One-to-One Functions

*Introduction to the unit; the three cases; vertical and horizontal line tests; reversibility*

### 1 Introduction to the unit

We now begin a new unit: **exponential functions and logarithmic functions**. Our goal in this lecture is to lay the groundwork for exponential functions, and for that the concept of a **one-to-one function** is very much relevant. The plan for the unit is the following.

**Plan for this unit**

1. The concept of a **one-to-one function** (this lecture).
2. The **definition of the exponential function**.
3. **Properties of exponential functions**, in the same spirit as the properties we studied for polynomials and for the straight line.
4. The **natural exponential function** — why it is called "natural", and the properties it shares with the rest of the mathematical world.

### 2 Recalling the concept of a function

Whenever we talk about a function we write

$$y = f(x),$$

and to specify $f$ we need a **domain**, say $A$, and a **co-domain**, say $B$:

$$f : A \longrightarrow B.$$

In general a function may be defined between any two sets. To simplify our understanding, and because we wish to work on the coordinate plane, we take

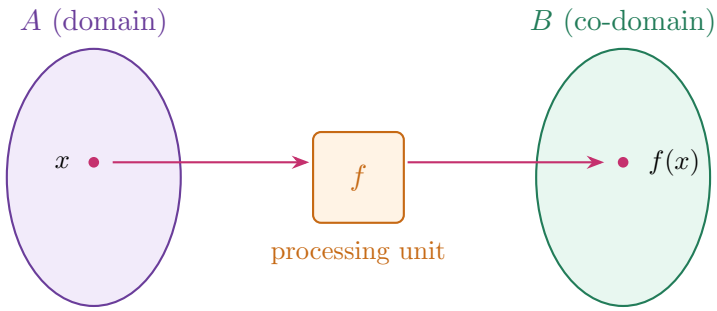
$$A \subseteq \mathbb{R}, \qquad B \subseteq \mathbb{R},$$

that is, $A$ and $B$ are subsets of the real line. Here $A$ is the domain and $B$ is the co-domain.

**What a function does**

A function is a **relation** between one set and another — a **mapping** that assigns to values of one set the values of the other set.

If we feed some value $x$ from the set $A$ into $f$, the machine processes it and gives out $f(x)$. It works like a popcorn machine: you feed in the corn and you get out the popcorn.




## 3 The three cases

Let us see what cases can possibly occur when we feed values into $f$.

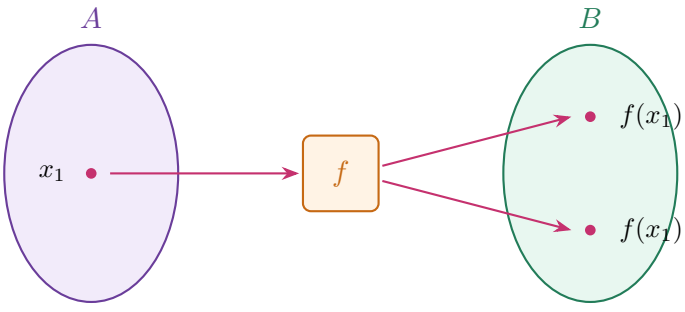
| Case | Description | Is it a function? | Is it reversible? |
|---|---|---|---|
| 1 | One $x$, more than one $f(x)$ | No (vertical line test fails) | — |
| 2 | More than one $x$, one $f(x)$ | Yes | No (horizontal line test fails) |
| 3 | One $x$, one $f(x)$ | Yes | Yes |

We now analyse the three cases one by one.

### 3.1 Case 1: one $x$, more than one $f(x)$

What do we mean by "one $x$, more than one $f(x)$"? Suppose we feed in the value $x_1$. One time it gives us a value $f(x_1)$, and another time it gives us a different value $f(x_1)$:

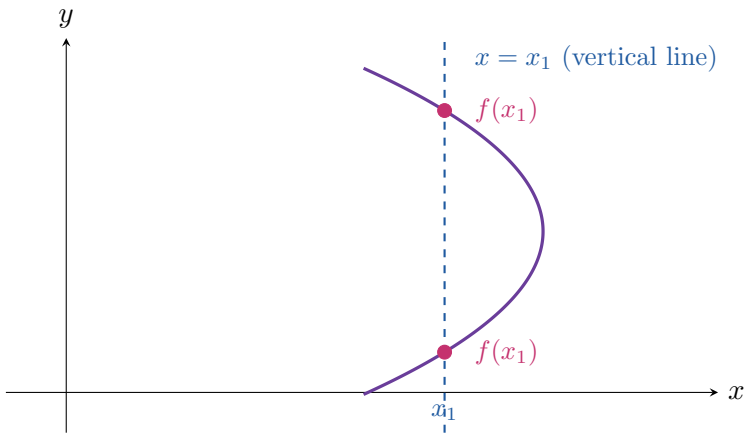
$$\text{same input } x_1 \longrightarrow \text{two outputs } f(x_1) \text{ that are not equal.}$$




two different outputs for the same input

Is this a function? Is it well behaved? **It is not.** If we feed in $x_1$ we do not know which value will come out, so this is **not a function**.

Since we are dealing with the coordinate plane, let us see the same situation there. Take a value $x_1$ on the $x$-axis. Once we feed it in we get one point $f(x_1)$, and another time we get a different point $f(x_1)$. If we draw a curve through such points, it looks as follows.




For the same value of $x$ we are getting two different values, so the function is not well behaved. This gives us a graphical criterion. Take the line

$$x = \text{constant}, \qquad \text{for example } x = x_1,$$

and draw it vertically — it is a line parallel to the $y$-axis. In the picture above this line intersects the graph at two points.

**Vertical line test**

Draw the vertical line $x =$ constant and slide it across the graph. If the line passes through more than one point of the graph, the **vertical line test fails** and the relation **cannot be a function**.

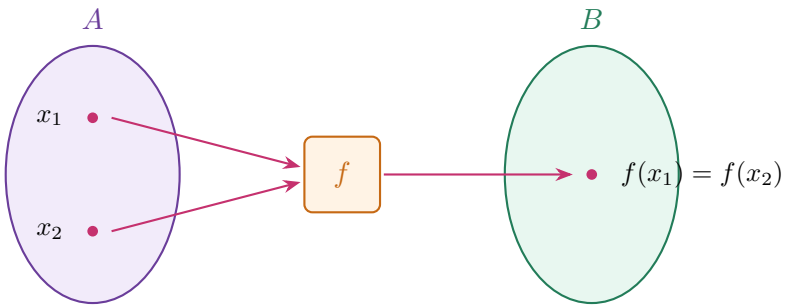
**Conclusion of Case 1**

The vertical line test fails, and hence this is **not a function**.

### 3.2 Case 2: more than one $x$, one $f(x)$

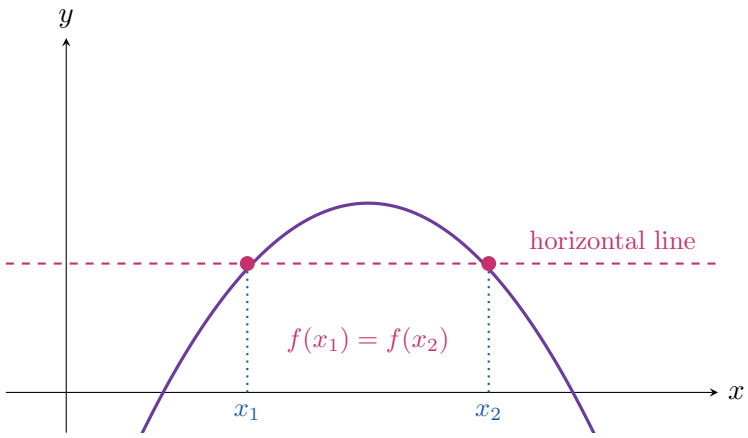
Here there are two different inputs $x_1$ and $x_2$, with

$$x_1 \neq x_2, \qquad \text{but} \qquad f(x_1) = f(x_2).$$




Is this a function or not? **The answer is yes, this is a function.** Two different inputs are allowed to be sent to the same output; what is forbidden is one input going to two outputs.

On the coordinate plane, we can join a curve passing through the two points $(x_1, f(x_1))$ and $(x_2, f(x_2))$, which lie at the same height.




Notice that if we rotate this graph by 90° and flip it over, we obtain a graph like the one in Case 1. This tells us that we are able to write $y$ as a function of $x$, but if we try to write $x$ as a function of $y$ — which is exactly the rotate-by-90°-and-flip picture — we cannot go back.

Concretely, suppose we set up another processing unit $g$ that is meant to undo $f$. Pick the value $y_1$ in the second set and feed it into $g$. Then one time $g$ gives $g(y_1) = x_1$, and another time $g(y_1) = x_2$:

$$g(y) = x_1 \quad \text{and} \quad g(y) = x_2, \qquad x_1 \neq x_2.$$

So $g$ itself is not a function — it is exactly the Case 1 situation. What we were trying to do was to **reverse** the function, and we can say plainly that this function is **not reversible**.

**Horizontal line test**

Draw a horizontal line and slide it across the graph. If it passes through more than one point of the graph, the **horizontal line test fails**, and the function is **not reversible**. The horizontal line test therefore detects whether a function is reversible or not.

**Conclusion of Case 2**

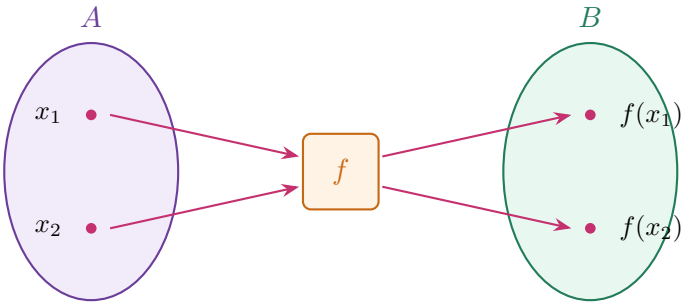
This **is** a function, but the horizontal line test fails, and hence it is **not reversible**.

### 3.3 Case 3: one $x$, one $f(x)$

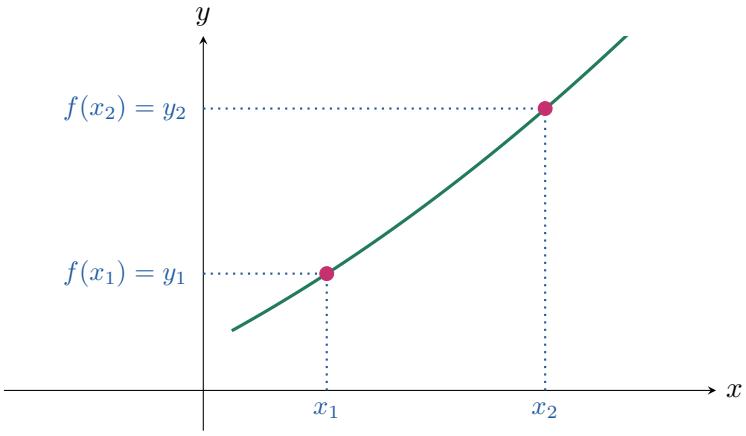
In the third case there is only one $x$ and one $f(x)$. If we substitute two different values $x_1$ and $x_2$ into the processing unit, we get $f(x_1)$ and $f(x_2)$, and these two are not equal to each other:

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2).$$

That is the only way "one $x$ to one $f(x)$" can happen: different inputs give different outputs.




What is the typical behaviour of such a function on the coordinate plane? Take $x_1$ and $x_2$: the corresponding heights $f(x_1)$ and $f(x_2)$ are different, and this happens for *every* pair of distinct inputs.




Is it a function? Yes — it is a properly defined function. And what can we say about reversibility? Since $x_1 \neq x_2$ gives $f(x_1) \neq f(x_2)$, if we start from

$$y_1 = f(x_1) \qquad \text{and} \qquad y_2 = f(x_2),$$

we can retrace our steps uniquely: there exists some function $g$ such that

$$x_1 = g(y_1), \qquad x_2 = g(y_2).$$

**Conclusion of Case 3**

This **is** a function, and it **is reversible**. Such a function is called a **one-to-one function**. A more precise word for "reversible" will be given in a later lecture.

## 4 Definition of a one-to-one function

The third case is the one we care about: one element in the domain, one element in the co-domain, and a clear-cut association between the two. Let us define it properly.

**Definition — One-to-one function**

A function $f$ is said to be **one-to-one** if, **for every** $x_1 \neq x_2$ belonging to the domain of the function,

$$\boxed{x_1 \neq x_2 \implies f(x_1) \neq f(x_2)}$$

There is another, equivalent, interpretation of this definition:

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

We shall use the first statement as our definition. However, sometimes it may be difficult to prove that form directly; in that case you may instead prove the one written in orange.

## 5 Summary of the three cases

- **Case 1 — one $x$, more than one $f(x)$:** not a function. Because we are dealing with the coordinate plane, the vertical line test fails — a vertical line passes through two or more points of the graph.

- **Case 2 — more than one $x$, one $f(x)$:** we were able to define a function properly, but we were not able to revert it, so the function is not reversible. This was decided on the basis of the horizontal line test, and it is related to Case 1: if $f$ is not reversible, then the reverse assignment $g$ is not a function either.
- **Case 3 — one $x$, one $f(x)$:** it is a function and it is reversible. This is the one-to-one function.

This brings us to the natural question: **how easy are one-to-one functions to identify?** That is exactly what we take up next.

# Lecture 2 — Examples of One-to-One Functions

*Vertical and horizontal line tests; increasing and decreasing functions*

## 1 The idea we are testing

Recall the definition we arrived at in Lecture 1, which we work with throughout this lecture.

**Definition 2.1 — One-to-one function**

A function $f : A \to B$ is said to be **one-to-one** if distinct inputs always produce distinct outputs, i.e.

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2), \qquad \text{for all } x_1, x_2 \in A.$$

Equivalently, in contrapositive form,

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

Before asking whether a given rule is one-to-one, we must first be sure that it is a function at all. The two graphical tests we use are the following.

## 2 Example 1: the modulus function $f(x) = |x|$

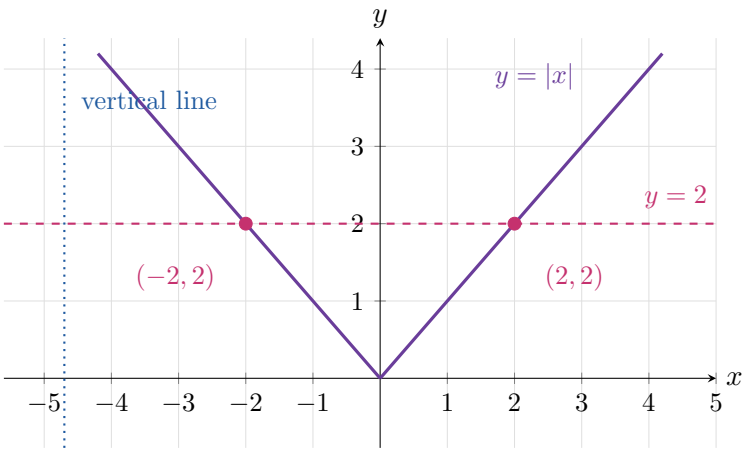
**Example 2.1**

Let $f(x) = |x|$. Is this function one-to-one?

Written out properly, the modulus function is the piecewise-defined function

$$f(x) = |x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

The branch $y = x$ is a straight line through the origin at 45°, and the branch $y = -x$ is its mirror image. Together they form a 90° "V" shape with its vertex at the origin.



### Step 1: Is it a function? (vertical line test)

Take a vertical line and slide it across the graph. At no position does it meet the V in more than one point. The vertical line test succeeds, and therefore $f(x) = |x|$ **is** a function.

### Step 2: Is it one-to-one? (horizontal line test)

Now pass a horizontal line through the graph, say the line $y = 2$. It cuts the graph at *two* points, $(-2, 2)$ and $(2, 2)$. So we have found $x_1 \neq x_2$ giving the same value of $f(x)$. Explicitly, take $x_1 = 2$ and $x_2 = -2$:

$$f(2) = |2| = 2, \qquad f(-2) = |-2| = 2 \qquad \Longrightarrow \qquad f(2) = f(-2), \quad 2 \neq -2.$$

**Conclusion of Example 2.1**

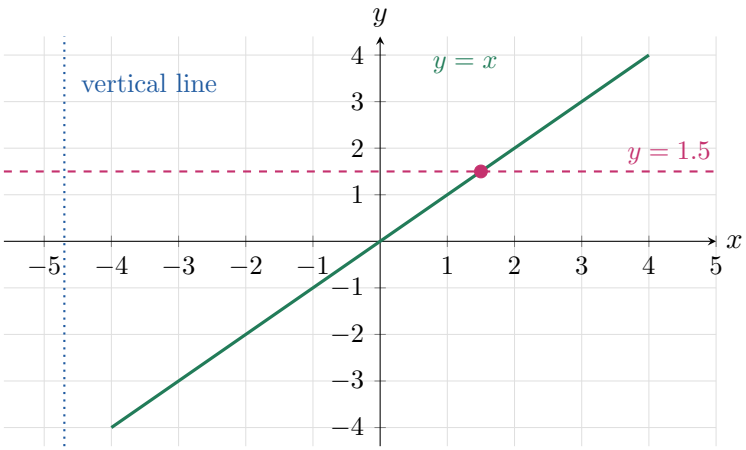
$f(x) = |x|$ is a valid function, but it is **not** a one-to-one function, since two distinct inputs 2 and $-2$ are mapped to the same output 2.

## 3 Example 2: the identity function $f(x) = x$

**Example 2.2**

Let $f(x) = x$. Is this function one-to-one?

The graph of $f(x) = x$ is a straight line passing through the origin.




**Vertical line test.** Start with the $y$-axis, slide it to the left and slide it to the right. Nowhere does the vertical line meet the graph in more than one point, so $f(x) = x$ is a valid function.

**Horizontal line test.** Now take a horizontal line and pass it from the top to the bottom. At every position it meets the line $y = x$ in exactly one point. So no two distinct inputs share an output, and the function is one-to-one.

Algebraically this is exactly what we expect:

$$x_1 \neq x_2 \implies f(x_1) = x_1 \neq x_2 = f(x_2).$$

**Conclusion of Example 2.2**

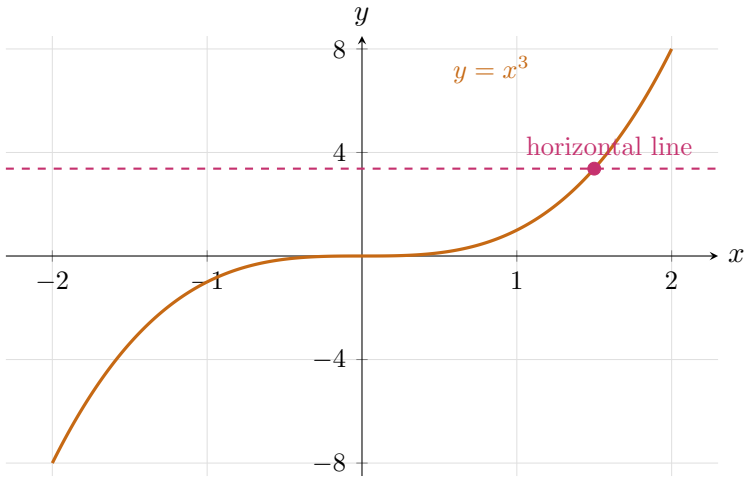
$f(x) = x$ **is** a one-to-one function.

## 4 Example 3: the cubic function $f(x) = x^3$

**Example 2.3**

Let $f(x) = x^3$. Is this function one-to-one?

The graph of $y = x^3$ rises through the origin, flattening near $x = 0$ and retaining its symmetry about the origin: it goes down on the left and up on the right.




Let the horizontal line travel up and down across the graph. At every height it meets the cubic curve in exactly one point — there is never a pair of points at the same level. Hence

$$x_1 \neq x_2 \implies f(x_1) = x_1^3 \neq x_2^3 = f(x_2).$$

**Conclusion of Example 2.3**

By the horizontal line test, $f(x) = x^3$ **is** a one-to-one function.

## 5 The horizontal line test as a theorem

**Theorem 2.1 — Horizontal line test**

If every horizontal line intersects the graph of a function $f$ in **at most one point**, then $f$ is one-to-one.

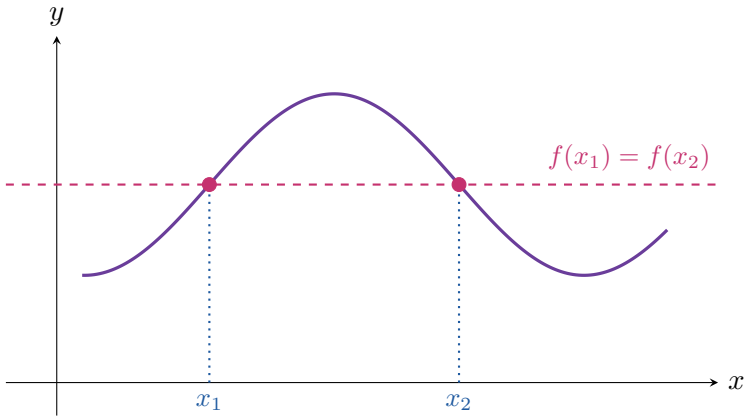
### Proof (by contraposition)

We show the contrapositive statement: *if $f$ is not one-to-one, then some horizontal line intersects the graph of $f$ in more than one point.*

Suppose $f$ is not one-to-one. Then by Definition 2.1 there exist two points $x_1 \neq x_2$ in the domain with

$$f(x_1) = f(x_2).$$

Mark $x_1$ and $x_2$ on the $x$-axis and mark the corresponding points on the graph. Since the two function values are equal, both graph points sit at the same height. The horizontal line drawn at that common height therefore passes through both of them, and so it meets the graph in more than one point.




This establishes the contrapositive, and hence the theorem: if a horizontal line intersects the graph of a function in at most one point, then $f$ is one-to-one. ■

## 6 A large class of one-to-one functions

Having a test is useful, but it is even more useful to recognise *whole classes* of functions that are one-to-one at a glance. Two such classes are the increasing and the decreasing functions.

**Definition 2.2 — Increasing function**

A function $f$ defined on a set $A$ is **(strictly) increasing** if

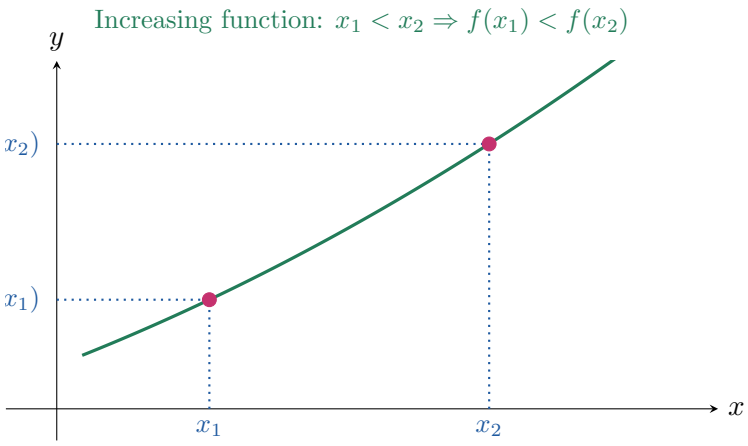
$$x_1 < x_2 \implies f(x_1) < f(x_2), \qquad \text{for every } x_1, x_2 \in A.$$

If we relax the strict inequalities we get the non-strict version,

$$x_1 \leq x_2 \implies f(x_1) \leq f(x_2),$$

but for the purpose of this lecture we use the *strict* form written in Definition 2.2.

The intuition is simple: if $x_1$ lies to the left of $x_2$ on the $x$-axis, then $f(x_1)$ lies below $f(x_2)$ on the $y$-axis.

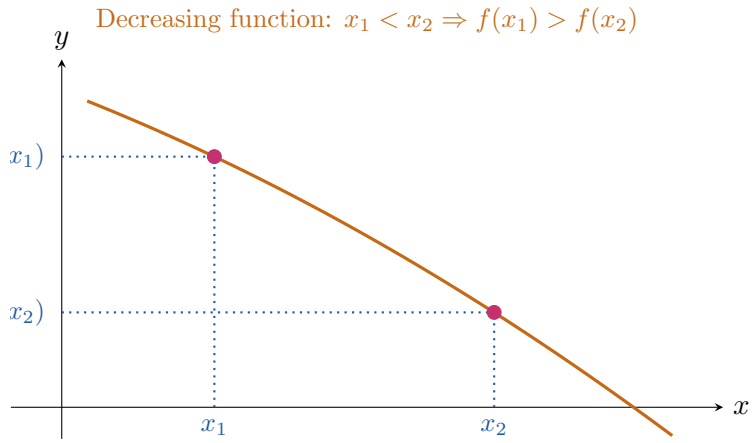



**Definition 2.3 — Decreasing function**

A function $f$ defined on a set $A$ is **(strictly) decreasing** if

$$x_1 < x_2 \implies f(x_1) > f(x_2), \qquad \text{for every } x_1, x_2 \in A.$$

A decreasing function is obtained, for instance, by multiplying an increasing function by $-1$. Here as we move from left to right, $x_1$ comes first and $x_2$ later, and the condition forces $f(x_1)$ to sit *above* $f(x_2)$.




**Theorem 2.2**

If $f$ is a strictly increasing function or a strictly decreasing function, then $f$ is one-to-one.

### Why this holds

Apply the horizontal line test to either graph. Roll a horizontal line up and down across the $y$-axis. For an increasing function the graph never comes back to a height it has already left; the same is true, in the opposite direction, for a decreasing function. So at no position does the horizontal line meet the graph in more than one point, and by Theorem 2.1 the function is one-to-one.

Equivalently, argue directly from the definitions. Take any $x_1 \neq x_2$. Then either $x_1 < x_2$ or $x_2 < x_1$, and in each case the defining inequality gives

$$f(x_1) < f(x_2) \quad \text{or} \quad f(x_1) > f(x_2),$$

so in particular $f(x_1) \neq f(x_2)$. Hence $f$ is one-to-one.

## 7 Summary of the examples

| Function | Is it a function? | Is it one-to-one? |
|---|---|---|
| $f(x) = \lvert x \rvert$ | Yes (vertical line test) | No — $f(2) = f(-2) = 2$ |
| $f(x) = x$ | Yes (vertical line test) | Yes — horizontal line test |
| $f(x) = x^3$ | Yes (vertical line test) | Yes — horizontal line test |
| Strictly Increasing $f$ | Yes | Yes — Theorem 2.2 |
| Strictly Decreasing $f$ | Yes | Yes — Theorem 2.2 |

## 8 Closing remark

One - one function carry an additional property that we will exploit next: they are **reversible**. With this insight we move on to the next topic, **exponential functions**.

# Lecture 3 — Exponential Functions

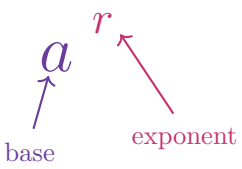
*Exponents for real powers; laws of exponents; the standard form $f(x) = a^x$*

## 1 Starting from a known idea: the exponent

Our next topic is the **exponential function**. Let us begin by identifying it with terminology we already know — the **exponent**. We have already seen exponents where integer powers were allowed, and later, while defining exponents, we allowed rationals as well. So far our exponents have been of the form

$$a^r, \qquad a > 0, \qquad r \in \mathbb{Q}.$$

As the name suggests, an exponential function has to do with the exponent. When we consider an expression of this form we are raising some number to a power:




Here $a$ is popularly called the **base** and $r$ is the **exponent**.

Now, if we want to define exponential functions on the whole real line, then it is mandatory for us to define $a^r$ for those $r$ that do *not* belong to the set of rational numbers, i.e. for

$$r \in \mathbb{R} \setminus \mathbb{Q},$$

which is the set of **irrational numbers**. As of now we do not know what the definition of such an exponent should be.

## 2 Why do we insist that $a > 0$?

We have been assuming $a > 0$ throughout. Let us see why. Suppose we take

$$a = -1$$

and suppose $a^{1/2}$ were defined — note that $\frac{1}{2}$ is a perfectly good rational exponent. Then

$$(-1)^{1/2} = i \in \mathbb{C},$$

a complex number. We do not want to deal with complex numbers here, so we avoid this by insisting on $a > 0$.

**Note**

In general one *can* define $a$ to be a negative number and then work with complex numbers, but we do not want to enter into that discussion. So $a < 0$ is eliminated, and henceforth $a > 0$ always.

## 3 Can we make sense of an irrational exponent?

With $a > 0$ settled, the remaining question is: what happens when $r$ is irrational, i.e. when $r \in \mathbb{R}\setminus\mathbb{Q}$? To be precise, let us ask

$$\text{Can we define} \quad 2^{\sqrt{2}} \quad \text{or} \quad 5^{\pi} \ ?$$

There is no direct way to answer this question, but there is a strategy — a calculus-based strategy.

### 3.1 The example $5^{\pi}$

We know that $\pi$ is an irrational number, and its numerical approximation is

$$\pi = 3.14159265\ldots$$

where the expansion is non-repeating and continues to infinity. From what we already know, can we define the number $5^{\pi}$? Not accurately, not yet. So let us ask a chain of easier questions:

| Question | Exponent as a fraction | Defined? |
|---|---|---|
| Is $5^{3}$ defined? | $3 = \dfrac{3}{1}$ | Yes |
| Is $5^{3.1}$ defined? | $3.1 = \dfrac{31}{10}$ | Yes |
| Is $5^{3.14}$ defined? | $3.14 = \dfrac{314}{100}$ | Yes |
| Is $5^{3.141}$ defined? | $3.141 = \dfrac{3141}{1000}$ | Yes |
| $\vdots$ | $\vdots$ | $\vdots$ |

Every one of these approximations is a **rational** number: 3 is rational, 3.1 is $\frac{31}{10}$, 3.14 is $\frac{314}{100}$, and we can go on with 3.141, and so on. Each of these powers is therefore already defined for us. If we continue this way, we will reach somewhere, and that somewhere is what we call

$$5^{\pi}.$$

$$5^{3} \longrightarrow 5^{3.1} \longrightarrow 5^{3.14} \longrightarrow 5^{3.141} \longrightarrow \cdots \longrightarrow 5^{\pi}$$

### 3.2 The exercise $2^{\sqrt{2}}$

In a similar manner, take

$$\sqrt{2} = 1.41\ldots$$

and build up the chain

$$2^{1} \longrightarrow 2^{1.4} \longrightarrow 2^{1.41} \longrightarrow \cdots \longrightarrow 2^{\sqrt{2}}.$$

Each of $2^{1}$, $2^{1.4}$, $2^{1.41}$, … is defined because each exponent is rational, and proceeding in this manner we reach $2^{\sqrt{2}}$.

**The platform for the exponential function**

In principle, therefore, we can define $a^{r}$ for an irrational number $r$. We conclude that

$$a^{x} \text{ is well defined for } x \in \mathbb{R},$$

and this is very important — it sets up the platform for defining an exponential function.

The justification comes from the **convergence of sequences**, a topic you will study later and one that is outside the scope of this syllabus. We know for sure that the value exists: the existence of $5^{\pi}$ is assured. If you are interested, you can take a basic course in analysis or calculus where these things are studied.

## 4 Laws of exponents

Let us now recall the simple laws of exponents. Earlier we knew these laws only for rational numbers; now we state them for **real** numbers. Throughout, $s$ and $t$ are real numbers playing the role of exponents, and $a$ and $b$ are positive numbers playing the role of bases:

$$s, t \in \mathbb{R}, \qquad a > 0,\ b > 0.$$

**Laws of exponents**

$$a^s\, a^t = a^{s+t} \qquad\qquad 1^s = 1, \quad \text{for every } s \in \mathbb{R}$$

$$(a^s)^t = a^{st} \qquad\qquad a^{-s} = \frac{1}{a^s} = \left(\frac{1}{a}\right)^s$$

$$(ab)^s = a^s\, b^s \qquad\qquad a^0 = 1$$

Two points are crucial in these laws:

- In $a^s\, a^t = a^{s+t}$, a **product** on the left becomes an **addition** in the exponent on the right.
- In $(a^s)^t = a^{st}$, the "raised to" operation becomes a **product** in the exponent.

Remember also that $a$ is always greater than 0 in $a^0 = 1$, because $0^0$ is **undefined**. These laws will be used left and right in what follows, so they are worth remembering thoroughly.

## 5 Definition of the exponential function

**Definition 3.1 — Exponential function (standard form)**

An **exponential function** in the standard form is given by

$$\boxed{f(x) = a^x, \qquad a > 0, \quad a \neq 1}$$

The condition $a \neq 1$ is newly introduced. We have already seen why $a > 0$; the condition $a \neq 1$ needs further analysis, which we take up below. So the permitted values of $a$ are all positive numbers, barring the values 0 and 1:




So we get two characterisations of the base:

$$0 < a < 1 \qquad \text{and} \qquad a > 1.$$

### 5.1 First observation: the domain

Because we have barred the values 0 and 1 and allowed every other positive base, the function

$$f(x) = a^x$$

has domain equal to the **entire real line**: for every $x \in \mathbb{R}$ we are able to compute $a^x$.

### 5.2 Second observation: why $a \neq 1$?

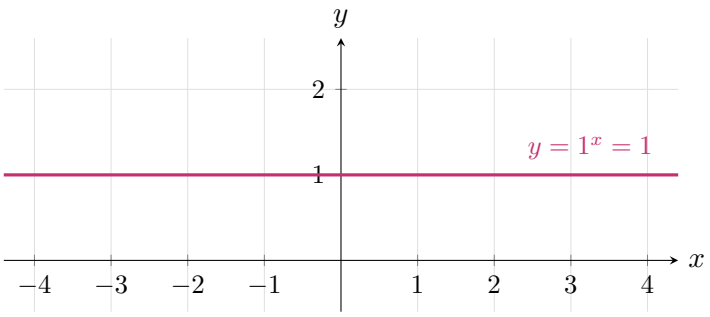
Let us put $a = 1$ and see what happens. Then

$$f(x) = 1^x.$$

But from the laws of exponents we know that $1^s = 1$ for every real $s$. Therefore

$$f(x) = 1^x = 1 \qquad \text{for all } x \in \mathbb{R},$$

which is nothing but a **constant function**, whose graph is the horizontal line $y = 1$.




We are not interested in handling a constant function, so we do not call this an exponential function — and that is exactly what the condition $a \neq 1$ in the definition says.

**Summary of the restrictions on the base**

Henceforth we will never talk about $a = 1$, $a = 0$, or a negative $a$. The base always satisfies

$$a > 0, \qquad a \neq 1,$$

which splits into the two cases $0 < a < 1$ and $a > 1$.

## 6 Exercise: explore the graphs

It will now be interesting to use a graphical tool and see what functions of this kind look like.

**Exercise 3.1**

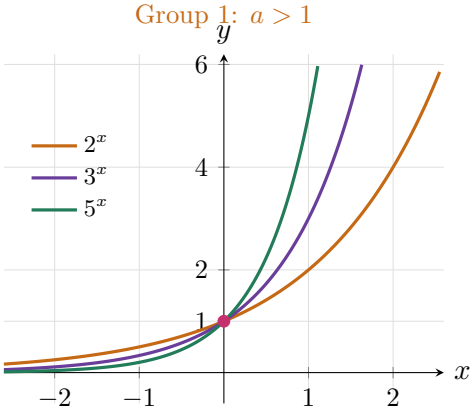
Use a graphing tool such as **Desmos** and plot the following functions.

1. Plot the first group of functions — those with base $a > 1$ — *together* on the same axes, without any prior understanding of the behaviour of the function.
2. Then plot the second group — those with base $0 < a < 1$ — together on the same axes.
3. **Identify the properties of the graphs:** through which points do they pass? Is there any difference between the graphs of group 1 and group 2?

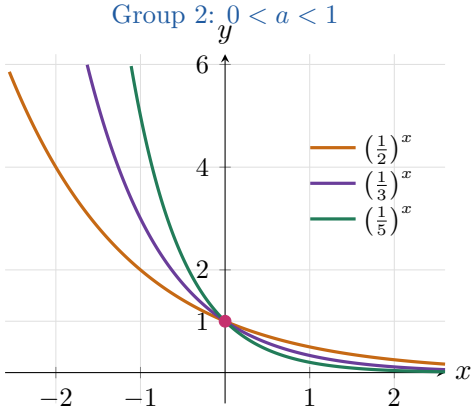
Identify all these properties in the same way we did for polynomials. After doing that, return to the next lecture, where we will look at some of these functions through their graphs and analyse them.

For reference while you experiment, here is the kind of picture the two groups produce. Plot your own before looking closely at these.

Group 1: $a > 1$




Group 2: $0 < a < 1$




Notice already the common point $(0, 1)$, which follows immediately from the law $a^0 = 1$.

# Lecture 4 — Graphs and Properties of Exponential Functions

*The prototypes $2^x$ and $\left(\frac{1}{5}\right)^x$; asymptotes, end behaviour, monotonicity*

## 1 Collecting our understanding through two examples

Having carried out the plotting exercise, let us recollect that understanding through two examples: one with base greater than 1, and one with base between 0 and 1.

## 2 Example 1: $f(x) = 2^x$

**Example 4.1**

Analyse the function $f(x) = 2^x$ completely — domain, range, intercepts, asymptote, end behaviour, roots and monotonicity — and then compare with its graph.

### 2.1 Domain

We have already discussed in detail that the domain of an exponential function is the **entire real line**:

$$\text{Domain of } f = \mathbb{R}.$$

### 2.2 Range

Look at $2^x$. Here the base 2 is greater than 1. Therefore, whenever $x$ is positive,

$$x > 0 \implies 2^x > 2^0 = 1.$$

And if $x$ is less than 0, then

$$x < 0 \implies 2^x < 1.$$

Can $2^x$ ever become negative? **No.** So it is always greater than 0:

$$2^x > 0 \qquad \text{for every } x \in \mathbb{R}.$$

With this understanding we can write down the range straight away:

$$\text{Range of } f = (0, \infty).$$

### 2.3 Intercepts

Something happens at the point $(0, 1)$: putting $x = 0$ gives $2^0 = 1$, so $(0, 1)$ is the **$y$-intercept**. Below that, for negative $x$, the graph goes below 1 — but it never goes below 0.

This is an interesting fact: since $2^x$ can never be a negative number, will the graph touch the $x$-axis? It will not. So

$$x\text{-intercept: nil.}$$

But it does *approach* 0: it never touches the line $y = 0$, yet it travels along it, touching it only "at infinity". Such a line is called a **horizontal asymptote**.

**Horizontal asymptote**

For $f(x) = 2^x$, the line

$$y = 0$$

is a **horizontal asymptote**: the graph approaches it arbitrarily closely without ever touching it.

### 2.4 End behaviour

As $x$ increases, $2^x$ also increases — in fact it increases at a more rapid rate than $x$ itself. And on the other side we have just identified $y = 0$ as the horizontal asymptote. So

$$x \to \infty \implies 2^x \to \infty, \qquad x \to -\infty \implies 2^x \to 0.$$

### 2.5 Roots

Does the function have any roots? From the graphical picture it is very clear that the curve never touches 0. Hence

there are no roots.

### 2.6 Increase and decrease

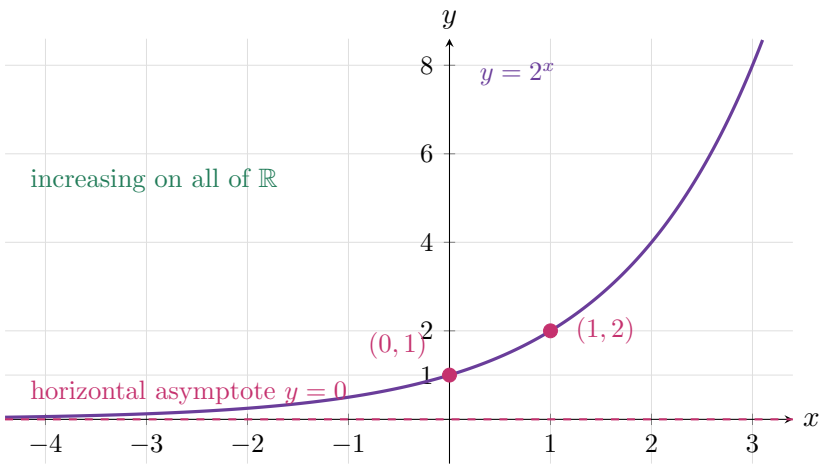
For polynomials we used to work out the domains of increase and decrease. Here there is no need to do so. Look at $2^x$ and take $x_1 \neq x_2$; without loss of generality let $x_1 < x_2$. What can we say about $2^{x_1}$ and $2^{x_2}$? Since $x_1 < x_2$, raising 2 to these powers preserves the relation:

$$x_1 < x_2 \implies 2^{x_1} < 2^{x_2}.$$

So the function is an **strictly increasing function** — and strictly increasing functions are **one-to-one**, as we established earlier. Throughout the real line the function is only increasing, so there is no doubt about domains of increase and decrease.

### 2.7 The graph

Let us now look at the graph and identify the special points. The first is the $y$-intercept $(0, 1)$. The second: at $x = 1$ the function takes the value 2, so $(1, 2)$ lies on the graph.



These two points are very special — they tell us something. Had the function been $a^x$ instead of $2^x$, the second point would have been

$$(1, a),$$

and since $a > 1$, that point lies above the level $y = 1$. As $x \to \infty$ the graph goes to infinity, and as $x \to -\infty$ the graph goes to 0. Reading from left to right, the curve rises: it is an increasing function, $y = 0$ is the horizontal asymptote, the range is $(0, \infty)$, and the domain is the entire real line.

## 3 The general case $a > 1$

What is so special about $2^x$? If we replace 2 by 3, we still get the $y$-intercept $(0, 1)$, because $3^0 = 1$ as well. We again get

$$\text{Domain} = \mathbb{R}, \quad \text{Range} = (0, \infty), \quad \text{no } x\text{-intercept},$$

$[y = 0]$ horizontal asymptote,

$$x \to \infty \Rightarrow a^x \to \infty, \qquad x \to -\infty \Rightarrow a^x \to 0, \qquad \text{no roots}, \qquad \text{only increasing}.$$

**Fact**

Every $f(x) = a^x$ with $a > 1$ has the **same properties** as $2^x$.

So there is no need to draw graphs for different values of $a$: the behaviour is the same, only the values change. In the graph above $(1, 2)$ was a point; if we consider $3^x$, then $(1, 3)$ is the corresponding point. The shape and every listed behaviour remain the same — only the values need to be evaluated. The general graph of $f(x) = a^x$ for $a > 1$ therefore looks exactly like the one above, with $(1, a)$ in place of $(1, 2)$.

Recall the line we drew for the base $a$, where we eliminated the two values 0 and 1. We have now settled the case $a > 1$; next we take up the case $0 < a < 1$.




## 4 Example 2: $g(x) = \left(\frac{1}{5}\right)^x$

**Example 4.2**

Take $g(x) = a^x$ with $0 < a < 1$, specifically

$$g(x) = \left(\frac{1}{5}\right)^x.$$

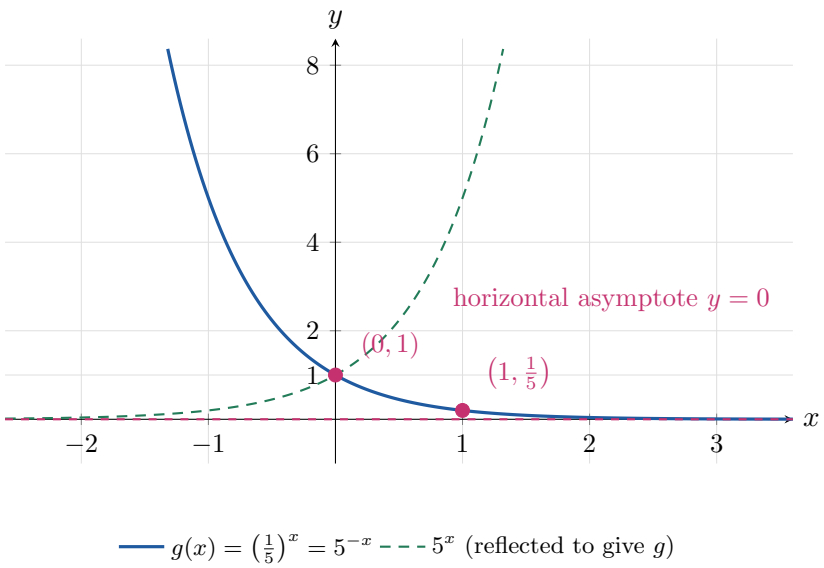
Analyse it in the same way.

We do not really have to draw this graph from scratch. Using the laws of exponents,

$$g(x) = \left(\frac{1}{5}\right)^x = 5^{-x},$$

so here $x$ has simply been replaced by $-x$. And we know what replacing $x$ by $-x$ does: it gives a **reflection across the $y$-axis** — something you have used in many assignment problems. So the

graph of $\left(\frac{1}{5}\right)^x$ is the mirror image of the graph of $5^x$ in the $y$-axis, coming down steeply from the left, flattening on the right, going very close to the $x$-axis but never touching it.




Even so, let us go through the analysis in the regular set-up.

- **Domain.** As we have used several times, the domain is the real line: Domain $= \mathbb{R}$.
- **Range.** Nothing changes, because a reflection across the $y$-axis does not alter the set of values: Range $= (0, \infty)$.
- **$y$-intercept.** A reflection across the $y$-axis leaves points on the $y$-axis fixed, so the $y$-intercept is still $(0, 1)$.
- **$x$-intercept.** Nil — and therefore **no roots**.
- **End behaviour.** Again by reflection: where $5^x$ went to infinity as $x \to \infty$, the reflected function does so as $x \to -\infty$; and where $5^x$ went to 0 as $x \to -\infty$, the reflected function does so as $x \to \infty$. Hence
$$x \to \infty \implies \left(\frac{1}{5}\right)^x \to 0, \qquad x \to -\infty \implies \left(\frac{1}{5}\right)^x \to \infty.$$
- **Monotonicity.** Because it is a reflection, the increasing behaviour becomes decreasing. There is no extra intelligence needed here: $g$ is a **strictly decreasing function**.

Now look at the given graph of $\left(\frac{1}{5}\right)^x$ and check that it coincides with our analysis. The point $(0, 1)$ is there. At $x = 1$ the value is $\frac{1}{5}$, so $\left(1, \frac{1}{5}\right)$ is on the graph. As $x \to \infty$ the function goes to 0; as $x \to -\infty$ it goes to infinity; and reading from left to right the curve comes down, so it is decreasing.

**Fact**

Every $f(x) = a^x$ with $0 < a < 1$ has the **same properties** as $\left(\frac{1}{5}\right)^x$. It is therefore a **representative class**: you do not have to worry about all the other functions, since they will all behave similarly.

## 5 Summary table

Let us summarise everything in a neat table. Given $f(x) = a^x$, the two zones on the base line — $0 < a < 1$ and $a > 1$ — behave as follows.

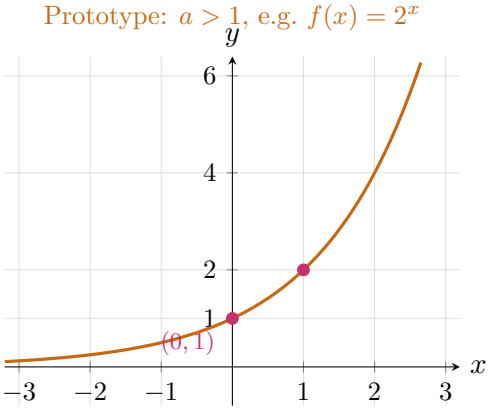
| Property | $0 < a < 1$ | $a > 1$ |
|---|---|---|
| Domain | $\mathbb{R}$ | $\mathbb{R}$ |
| Range | $(0, \infty)$ | $(0, \infty)$ |
| $x$-intercept | Nil | Nil |
| $y$-intercept | $(0, 1)$ | $(0, 1)$ |
| Roots | None | None |
| Horizontal asymptote | $y = 0$ | $y = 0$ |
| Monotonicity | Decreasing | Increasing |
| End behaviour ($x \to \infty$) | $f(x) \to 0$ | $f(x) \to \infty$ |
| End behaviour ($x \to -\infty$) | $f(x) \to \infty$ | $f(x) \to 0$ |

The only distinguishing feature is that the function is increasing in one zone and decreasing in the other; and because of that, the end behaviour changes accordingly. Being decreasing, it decreases to 0 — it is bounded below by 0; being increasing, it increases to infinity on the other side.

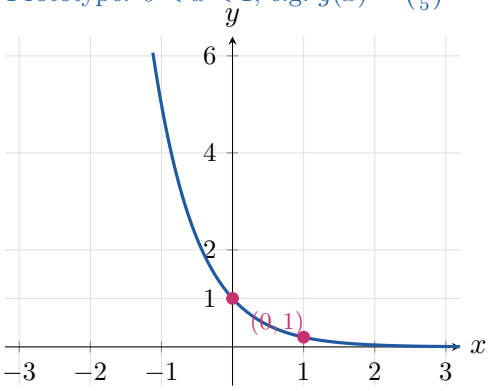
## 6 The two prototypes

Finally, look at the graphs of the two prototype functions side by side.

Prototype: $a > 1$, e.g. $f(x) = 2^x$




Prototype: $0 < a < 1$, e.g. $g(x) = \left(\frac{1}{5}\right)^x$




This completes our topic on exponential functions. Next we introduce something called the **natural exponential function**.

# Lecture 5 — The Natural Exponential Function

*The number $e$; continuous compounding; $f(x) = e^x$ and its special properties*

## 1 Where the number $e$ comes from

Among all exponential functions there is one more that is especially important: the **natural exponential function**. Its theory is derived from calculus, and to understand how it arises we need the theory of **limits**. In particular, this function depends on raising something to an exponent, where that something is an irrational number called $e$.

From the theory of limits it is known that

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$$

The existence of $e$ is studied in calculus

**Facts about $e$**

- $e$ is an **irrational number**.
- $e \approx 2.71828\ldots$

## 2 Why is $e$ so important?

### 2.1 The limit, seen numerically

When we say that $n$ grows without bound and

$$f(n) = \left(1+\frac{1}{n}\right)^n$$

converges to $e$, what do we mean? Let us put it formally with a table of values.

**Values of $\left(1+\frac{1}{n}\right)^n$**

| $n$ | $\left(1+\frac{1}{n}\right)^n$ |
|---|---|
| 1 | 2 |
| 10 | 2.5937 |
| 100 | 2.7048 |
| 1000 | 2.7169 |
| 10 000 | 2.7181 |
| 1 00 000 | 2.7183 |

At $n = 1$ the number is simply 2. At $n = 10$ it becomes 2.5937. Does this mean the function grows without bound? **No** — that is exactly why we get convergence, and questions of this type are studied in calculus. Substituting further, $n = 100$ gives 2.7048 and $n = 1000$ gives 2.7169: we

are getting closer to the ideal value of $e$. At $n = 10\,000$ we get 2.7181, and at $n = 1\,00\,000$ we get 2.7183, which *agrees with $e$ to four decimal places* — since $e = 2.71828\ldots$ also rounds to 2.7183. Going further — to a million and beyond — gives further improvement, but since we are focusing on four digits after the decimal, this much calculation is enough for us.

## 2.2 Continuous compounding

Another setting in which $e$ becomes crucial is interest calculation. You must have heard the term **continuous compounding**; it means taking exponents with respect to $e$. Let us demonstrate.

Suppose you invest Rs. 1 in a bank offering an interest rate of 1% for 1 year. Then the amount is

$$\left(1 + \frac{1}{100}\right)^1 = 1.01,$$

that is, on Rs. 1 you get 1 paisa of interest. Continuing this way for $n$ years,

$$\left(1 + \frac{1}{100}\right)^n.$$

But banks do not credit interest annually — they credit it **quarterly**. The interest rate is computed on a quarter, and whatever interest has accumulated is taken into account for the next quarter. So the bank takes the interest rate 0.01 and divides it into 4 parts:

$$\left(1 + \frac{0.01}{4}\right)^4.$$

If the bank instead revises the interest rate every **month**, then for a single year the same logic gives

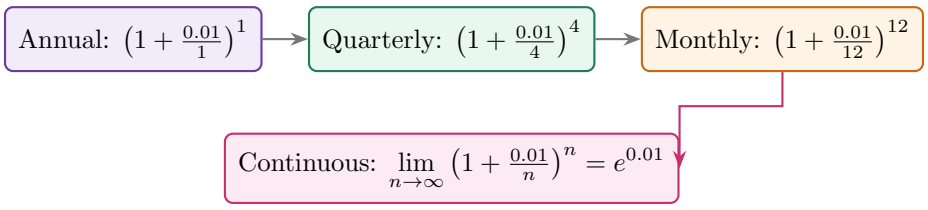
$$\left(1 + \frac{0.01}{12}\right)^{12}.$$

Now, if the bank starts revising the interest rate **infinitely often**, we are talking about

$$\lim_{n\to\infty} \left(1 + \frac{0.01}{n}\right)^n.$$

By the same judgement as before — where the numerator was 1 and now it is 0.01 — this converges to

$$e^{0.01}.$$




This is an interesting revelation: if you invest Rs. 1, then multiplying it by $e^{0.01}$ gives the interest accumulated along with the original capital, provided the bank follows continuous compounding. This is how interest is calculated whenever you study finance. That was for a period of one year; adding the period in terms of time $t$,

$$\text{Rs. } 1 \longrightarrow e^{0.01t}.$$

Replacing the 1% by a generic number $x$, the same discussion gives

$$\left(1 + \frac{x}{n}\right)^n \xrightarrow[n\to\infty]{} e^x, \qquad \left(1 + \frac{x}{n}\right)^{nt} \xrightarrow[n\to\infty]{} e^{xt}.$$

So $e$ typically arises whenever we consider continuous compounding.

### 2.3 Why "natural"?

The number $e$ comes up naturally in continuous compounding, and it also arises very naturally in the field of **differential equations**, which again relies on calculus. In that area the number has a special name — **Euler's number** — and it is worth searching for its meaning and relevance. That is how $e^x$ comes to be called the *natural* exponential function.

## 3 Definition and properties

**Definition 5.1 — Natural exponential function**

The **natural exponential function** is defined as

$$\boxed{f(x) = e^x}$$

What are the interesting properties of this function? They are very similar to those of the general exponential function we have studied, but it is special in some ways.

- **Domain:** $\mathbb{R}$.
- **Range:** $(0, \infty)$, the positive real line.
- Since $e \approx 2.71828$, naturally $e > 1$, so the graph is of the $a > 1$ type studied earlier.
- It passes through the two typical points $(0, 1)$ and $(1, e)$.
- **End behaviour:** as $x \to \infty$ the function goes without bounds; as $x \to -\infty$ it asymptotically approaches the $x$-axis.
- **Horizontal asymptote:** $y = 0$.
- It is an **strictly increasing** function.

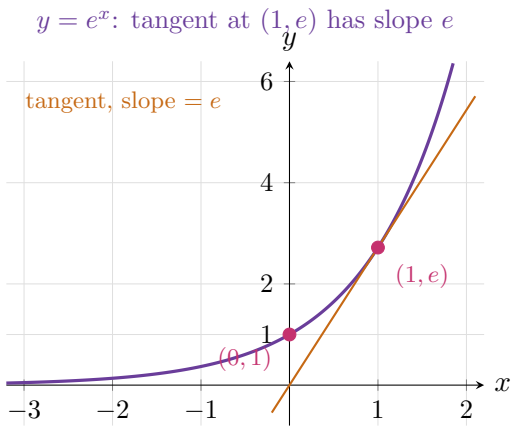
## 4 What makes $e$ special

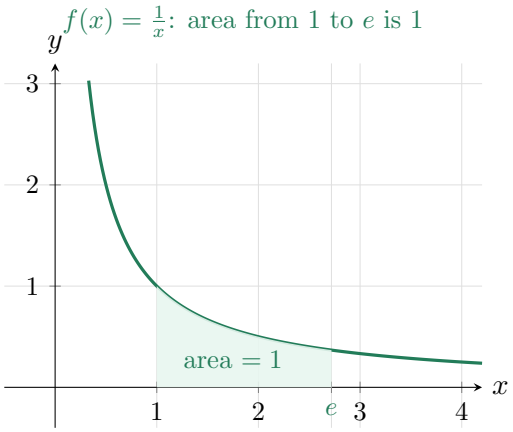
The properties above hold for a general exponential function too. Here are three things that are special to $e$ and are *not* true in general.

**Three special properties of the natural exponential function**

1. The **slope** of the line tangent to the curve $y = e^x$ at the point $(1, e)$ is exactly $e$.
2. The **area** covered under the curve $y = e^x$ from $-\infty$ to 1 is exactly $e$.
3. For the function $f(x) = \dfrac{1}{x}$, the **area under the curve between** $x = 1$ **and** $x = e$ is exactly 1 — a unit area, even though $e$ is an irrational number.

Why these hold is a matter for calculus to explore




## 5 Example: response to affiliate links

**Example 5.1**

Let $R$ be the percentage of people who respond to affiliate links under YouTube descriptions and purchase the product within $t$ minutes. Based on the available statistics, the function is

$$R(t) = 50 - 100\,e^{-0.2t}.$$

Answer the following.

(a) What percentage of people respond after 10 minutes?

(b) What is the highest percentage expected?

(c) How long before $R(t)$ exceeds 30%?

**Setting.** When you watch a video on YouTube, the speaker may mention affiliate links in the description. If you click such a link and purchase the product, the channel owner earns a commission. The person providing the links is interested in how many people respond within $t$ minutes, and has derived the function $R(t)$ above from the available data. We take the function as it is. A response rate of 30% is already good enough for merely placing some affiliate links.

### (a) Percentage responding after 10 minutes

We simply evaluate $R(10)$:

$$R(10) = 50 - 100\,e^{-0.2\times 10} = 50 - 100\,e^{-2}.$$

Computing $e^{-2}$ with a calculator,

$$R(10) \approx 36.47\ \%.$$

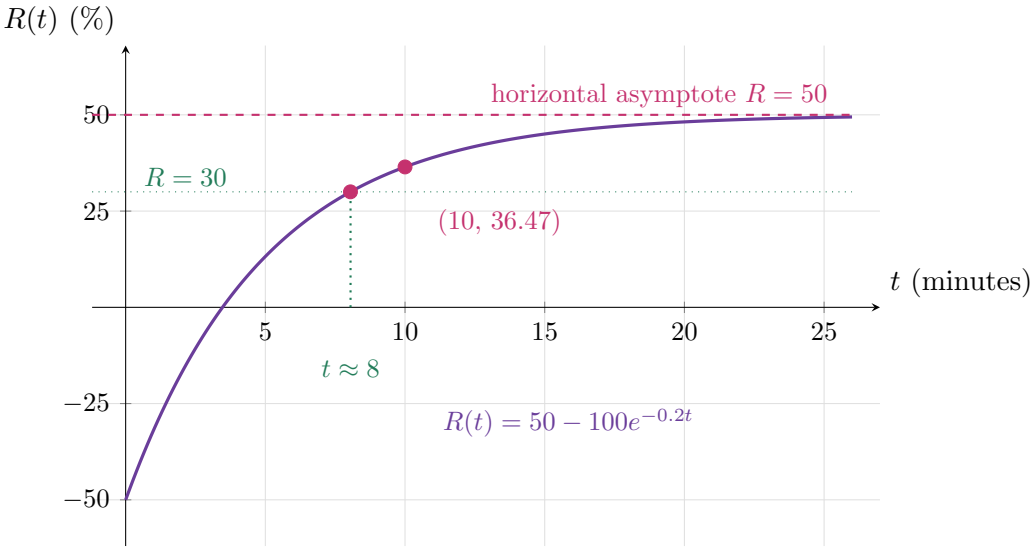
### (b) The highest percentage expected

Think about the function $50 - 100\,e^{-0.2t}$ by building it up from $e^x$ through transformations.

**Building the graph by transformations**

1. Start with $f(x) = e^x$, whose graph we know.
2. Replacing $x$ by $-x$ gives $e^{-x}$: a **reflection across the $y$-axis**. It never touches the $x$-axis but now falls from left to right.

3. Multiplying by $-100$ reflects the graph **across the $x$-axis** and scales it.
4. Adding 50 **shifts the graph up** by 50 units, which also moves the horizontal asymptote up to $y = 50$.




From the graph the curve asymptotically reaches 50 units, so the highest percentage expected is **50**% — it will not exceed 50%.

We can also argue without the graph. The function $e^{-0.2t}$ never exceeds 1, and as $t \to \infty$,

$$e^{-0.2t} \to 0.$$

So whether we multiply by 100 or by 10 000, the whole term $100\,e^{-0.2t}$ goes to 0 as $t \to \infty$, and therefore 50 is the maximum that can be achieved.

### (c) How long before $R(t)$ exceeds 30%

Formally, we set $R(t) = 30$ and solve for $t$:

$$\begin{aligned} 30 &= 50 - 100\,e^{-0.2t} \\ 30 - 50 &= -100\,e^{-0.2t} \\ -20 &= -100\,e^{-0.2t} \\ \frac{20}{100} &= e^{-0.2t} \\ \frac{1}{5} &= e^{-0.2t}. \end{aligned}$$

Here we must stop: at present we have no way of extracting $t$ from $e^{-0.2t} = \frac{1}{5}$ — no analytical route is available to us yet. If we could somehow write $t =$ something, the question would be answered.

So let us compute it **graphically** instead. The line $R(t) = 30$ meets the curve at a point; dropping from that point onto the horizontal axis — which measures $t$ — and reading off the mesh, the value turns out to be roughly 8. Hence

$$t \approx 8 \text{ minutes}.$$

This is how we can solve an expression involving exponential functions graphically, without solving it analytically at all. In the next lecture we will see how to solve this particular problem properly.

# Lecture 6 — Composite Functions

*A shopping example; the notation $f \circ g$; the domain of a composition*

## 1 Motivation: two offers at a computer store

Suppose it is known that you are a very good bargainer, and your friend wants to buy a computer. Your friend takes you to a computer store where all items are on sale, and two offers are available:

- **Offer 1:** you get the product at 85% of the price (i.e. a 15% discount).
- **Offer 2:** you get a flat Rs. 3000 off on the MRP (maximum retail price).

Being a good bargainer, you negotiate with the salesperson and strike a deal: the computer will be given at 85% of the price, *and* once that 85% amount is decided, a further Rs. 3000 will be given as a discount on top of it.

When written mathematically, this kind of thing is a **composite function** — and in fact you use such tricks in day-to-day life all the time. Let us see how the composition arises.

## 2 Writing the offers as functions

For cleanliness, let

$$x = \text{the item price (MRP)}.$$

The first offer gives 85% of the price, so it can be written as

$$f(x) = 0.85\,x.$$

The second offer subtracts Rs. 3000 from the MRP, so

$$g(x) = x - 3000.$$

**Implicit assumptions**

When a store offers such deals, it is safe to assume that

- no item on sale costs less than Rs. 3000, so $x > 3000$; and
- the store has already made sure it never has to pay money back, so after taking 85% of the price the amount is still more than 3000, i.e. $0.85x > 3000$.

These conditions are assumed implicitly here; we deal with them properly when formulating the problem in general.

## 3 The deal you struck, as a composition

The offer you actually obtained is 85% of the price, minus 3000. Writing it as a function,

$$h(x) = 0.85\,x - 3000.$$

When dealing with functions in mathematics, it is good to ask whether $h$ has any correspondence with $f$ and $g$. That is exactly the question composite functions answer.

Look at what is being done. Since $f(x) = 0.85x$, we may substitute $f(x)$ for that term:

$$h(x) = f(x) - 3000.$$

Is that a safe thing to do? Yes, of course — $f(x)$ *is* $0.85x$, so it is a perfectly valid substitution.

Now treat $f(x)$ as a single argument, just like $x$. The function $g$ takes whatever it is given and subtracts 3000:

$$g(\square) = \square - 3000.$$

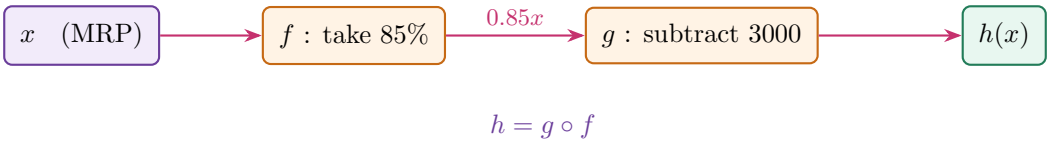
Right now what sits in the box is $f(x)$, so

$$g\big(f(x)\big) = f(x) - 3000 = 0.85x - 3000.$$

Therefore

$$\boxed{h(x) = g\big(f(x)\big) = (g \circ f)(x)}$$

In mathematics we write this as $g \circ f$, read as "$g$ composed with $f$". So our $h$ can also be written in terms of $g$ and $f$ in this fashion.




## 4 Evaluating a composite function

**Example 6.1**

Take $x = 14000$ and compute $(g \circ f)(x)$.

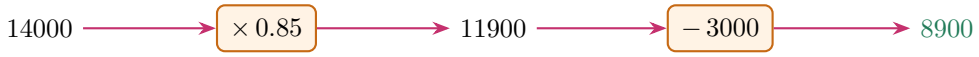
It is very simple — follow the same notion as above:

$$\begin{aligned}(g \circ f)(x) &= g\big(f(x)\big) \\ &= f(x) - 3000 \\ &= 0.85\,x - 3000.\end{aligned}$$

Now substitute $x = 14000$:

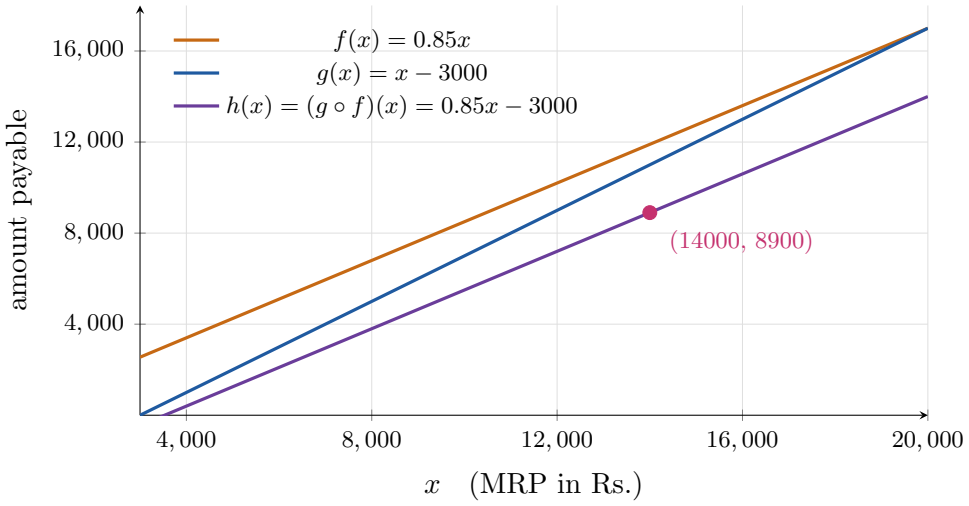
$$\begin{aligned}(g \circ f)(14000) &= 0.85 \times 14000 - 3000 \\ &= 11900 - 3000 \\ &= 8900.\end{aligned}$$

So the final answer is Rs. 8900.




What we have just shown is the evaluation of the composite function $g \circ f$, which is our $h$. There is nothing special about it — it is just the nomenclature we use — but this kind of composition helps in understanding a great many things.

Here is the picture of the three functions involved, over the price range where the offers apply.




## 5 Formal definition

A composition of functions is again a function, so natural questions about domain and range arise. We will answer them as they come. First, the definition.

**Definition 6.1 — Composition of functions**

The **composition of the functions $f$ and $g$** — also read as the composition of the function $f$ with $g$ — is denoted by $f \circ g$ and is defined by

$$\boxed{(f \circ g)(x) = f\big(g(x)\big)}$$

Note that at least two functions are needed to form a composition.

## 6 Domain of the composite function

**Definition 6.2 — Domain of $f \circ g$**

The domain of the composite function $f \circ g$ is the set of all $x$ such that

1. $x$ is in the domain of $g$; **and**
2. $g(x)$ is in the domain of $f$.

Why are these two conditions required? Focus on the expression $f\big(g(x)\big)$.

For something to lie in the domain of $f \circ g$, the expression must be **well defined**: when we input the value it must give us an output. If there is any ambiguity, it is not a properly defined function.

**Condition 1: $x \in \mathrm{Dom}(g)$**

What if $x$ is not in the domain of $g$? Then $g(x)$ is simply not defined, because $g$ is defined only over its own domain. So $x$ must be in the domain of $g$.

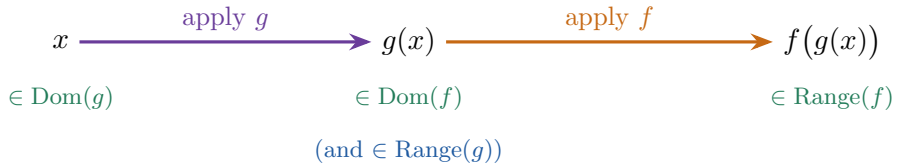
**Condition 2: $g(x) \in \mathrm{Dom}(f)$**

In the composite function we apply $f$ to the value obtained by applying $g$ — it is $g(x)$ that plays the part of the input to $f$. So if that value $g(x)$ is not in the domain of $f$, then $f\big(g(x)\big)$

is again not defined. Hence $g(x)$ must also lie in the domain of $f$.

## 6.1 Visualising the two conditions

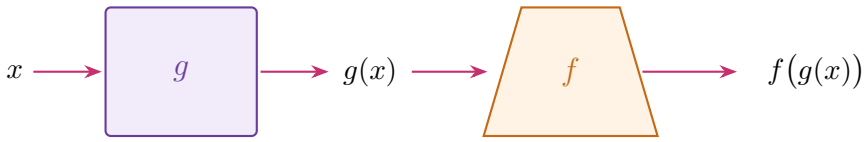
We can visualise it this way: $x$ is mapped by $g$ to the value $g(x)$, and that value is then fed to $f$ to produce $f\big(g(x)\big)$.




Here Dom is our abbreviation for domain. The middle value $g(x)$ is required to sit inside the domain of $f$; note that the collection of such values need not cover the *entire* domain of $f$ — it can be smaller. The final output belongs to the range of $f$.

## 6.2 The machine picture

To visualise it in a better way, think of **function machines**. You feed the input $x$ into the box $g$, and it spits out $g(x)$. That output is then fed into a second machine $f$.




must be in Dom($f$), else the machine cannot produce an output

# Lecture 7 — Practice with Composite Functions

*Computing $g \circ f$ and $f \circ g$; two working methods; an exercise*

## 1 Getting some practice

We have understood, roughly, the theory behind composite functions — the composition of two functions. It is now time to get some practice, so let us start with an example.

## 2 Worked example

**Example 7.1**

You are given the two functions

$$f(x) = 3x - 4, \qquad g(x) = x^2.$$

Find

(a) $(g \circ f)(x)$, and

(b) $(f \circ g)(x)$.

### 2.1 Part (a): computing $(g \circ f)(x)$

As per our theory, we first write

$$(g \circ f)(x) = g\big(f(x)\big).$$

**Method 1 — treat the argument as a box**

Go to the function $g$ and treat it as acting on a box: whatever sits inside the box gets squared,

$$g(\square) = \square^2.$$

Right now the box holds the argument $f(x)$, so we simply write

$$g\big(f(x)\big) = \big[f(x)\big]^2.$$

The entire process is now simplified — we no longer have to worry about what $g$ is. Substituting $f(x) = 3x - 4$,

$$(g \circ f)(x) = (3x - 4)^2.$$

**Method 2 — substitute the value of $f(x)$ first**

Alternatively, feed in the value of $f(x)$ straight away:

$$(g \circ f)(x) = g\big(f(x)\big) = g(3x - 4),$$

and since $g$ squares its argument,

$$g(3x - 4) = (3x - 4)^2.$$

### 2.2 Part (b): computing $(f \circ g)(x)$

Again there is no question about the first step:

$$(f \circ g)(x) = f\big(g(x)\big).$$

Here $f$ is the outer function, and $f(x) = 3x - 4$. To be very clear and avoid any confusion, write $f$ as acting on a symbol — say a triangle:

$$f(\triangle) = 3\,\triangle - 4.$$

Now the triangle is replaced by $g(x)$, that is all:

$$f\big(g(x)\big) = 3\,g(x) - 4.$$

But what is $g(x)$? Going back to the question, $g(x) = x^2$. Substituting,

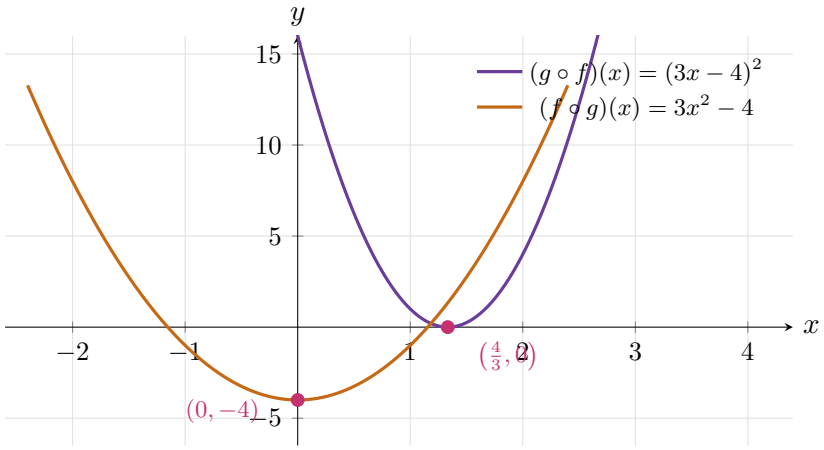
$$(f \circ g)(x) = 3x^2 - 4.$$

| Composition | Outer function acts on | Answer |
|---|---|---|
| $(g \circ f)(x) = g\big(f(x)\big)$ | $\big[f(x)\big]^2$ | $(3x-4)^2$ |
| $(f \circ g)(x) = f\big(g(x)\big)$ | $3\,g(x) - 4$ | $3x^2 - 4$ |

So we have seen how to write the compositions both ways, $g \circ f$ and $f \circ g$. Notice that the two answers are *not* the same:

$$(3x-4)^2 \;\neq\; 3x^2 - 4.$$

Their graphs make the difference plain.




## 3 Exercise

**Exercise 7.1**

Let

$$f(x) = x + 1, \qquad g(x) = x^2 - 1.$$

Find $(g \circ f)(x)$ and $(f \circ g)(x)$.

Pause here, work the exercise out on your own using either of the two methods above, and only then look at the answers below. It is good practice for revising the concepts.

**Check your answers**

$$(g \circ f)(x) = g\big(f(x)\big) = \big[f(x)\big]^2 - 1 = (x+1)^2 - 1 = x^2 + 2x,$$
$$(f \circ g)(x) = f\big(g(x)\big) = g(x) + 1 = \big(x^2 - 1\big) + 1 = x^2.$$

# Lecture 8 — Domain of a Composite Function

*Two exclusion rules; a worked example; a verification exercise*

## 1 The question

Let us go further and talk about how to **determine the domain of a composite function**. This is an important question. We are considering

$$(f \circ g)(x) = f\big(g(x)\big),$$

where all functions involved are real valued.

Recall the domain of a function, from Week 1: it is the set of allowed values for which the function is **well defined** — whatever input values you feed in, the function must produce a proper output.

## 2 The two rules

In order to determine the domain there are rules that must be followed, and consequently certain values must be **excluded** from the input values of $x$.

This is in concordance with what we saw earlier, where the two conditions were $x \in \mathrm{Dom}(g)$ and $g(x) \in \mathrm{Dom}(f)$. Earlier we were looking at which values are *possible*; now we look at the same thing from the other side — which values must be *excluded*.

**Rule 1**

$$x \notin \mathrm{Dom}(g) \implies x \notin \mathrm{Dom}(f \circ g).$$

If $x$ is not in the domain of $g$, then it cannot be included in the domain of $f \circ g$.

**Order matters**

The rule above is for $f \circ g$. When you talk about $g \circ f$, the corresponding statement is

$$x \notin \mathrm{Dom}(f) \implies x \notin \mathrm{Dom}(g \circ f).$$

So remember the order in which the functions are taken — it matters.

**Rule 2**

The set

$$\big\{\, x \;:\; g(x) \notin \mathrm{Dom}(f) \,\big\}$$

must **not** be included in the domain of $f \circ g$.

Otherwise we will have ambiguity; to eliminate that ambiguity these two rules must be followed very strictly. Let us now see, through an example, how things fail when they are ignored.

## 3 Worked example

**Example 8.1**

Given

$$f(x) = \frac{2}{x-1}, \qquad g(x) = \frac{3}{x},$$

find $(f \circ g)(x)$ and the domain of $f \circ g$.

### 3.1 Finding $(f \circ g)(x)$

Simply apply the definition — there is no confusion here:

$$(f \circ g)(x) = f\big(g(x)\big).$$

Using the box form of $f$,

$$f(\square) = \frac{2}{\square - 1} \qquad \Longrightarrow \qquad f\big(g(x)\big) = \frac{2}{g(x) - 1}.$$

Now substitute $g(x) = \dfrac{3}{x}$:

$$(f \circ g)(x) = \frac{2}{\dfrac{3}{x} - 1}.$$

Assuming $x \neq 0$, multiply numerator and denominator by $x$ and simplify:

$$\boxed{(f \circ g)(x) = \frac{2x}{3-x}}$$

### 3.2 Finding the domain

Looking at the simplified expression, we can see immediately that at $x = 3$ the function is not defined, because the denominator becomes 0 — and $\frac{6}{0}$ is undefined. So the domain must exclude 3. That much is well known. But let us now check, because of the composition, whether any further points must be eliminated.

#### Applying Rule 1

Look at $g(x) = \dfrac{3}{x}$. This function is well defined only when $x \neq 0$, so

$$x = 0 \notin \mathrm{Dom}(g) \implies x = 0 \notin \mathrm{Dom}(f \circ g).$$

**A tempting — but wrong — argument**

You may object: looking at the simplified form $\dfrac{2x}{3-x}$ and substituting $x = 0$ gives $\dfrac{0}{3} = 0$, so surely the function is well defined there?
No. While arriving at that simplified form we *multiplied numerator and denominator by* $x$, assuming $x \neq 0$. Had we not assumed $x \neq 0$, we would never have reached that form. So $x \neq 0$ remains a valid condition even though nothing appears wrong in the simplified expression — we are composing two functions, and $x = 0$ is outside the domain of the inner one.

### Applying Rule 2

Look at $f(x) = \dfrac{2}{x-1}$: it is well defined only when $x \neq 1$. So we need $g(x) \neq 1$:

$$g(x) = \frac{3}{x} = 1 \quad \Longrightarrow \quad x = 3.$$

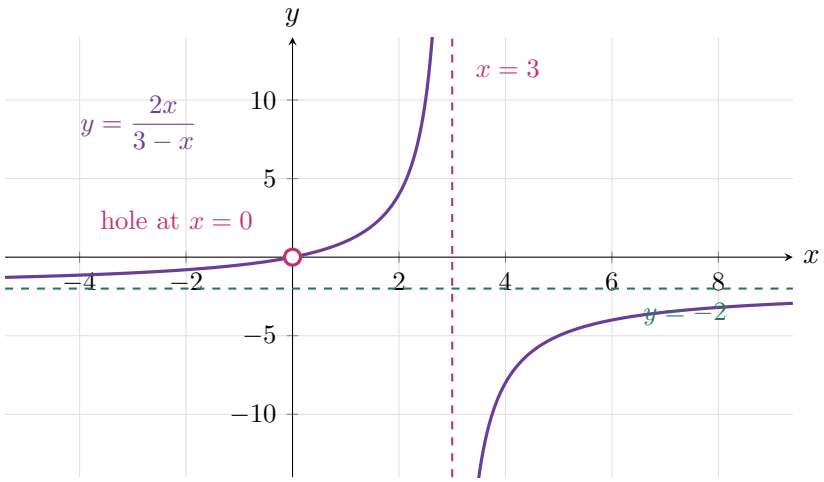
Hence $x = 3$ must also be eliminated from the domain of $f \circ g$ — which agrees with what we observed directly from the simplified expression.

### Conclusion

At every other point both $f$ and $g$ are well defined. Therefore

$$\mathrm{Dom}(f \circ g) \;=\; \{\, x \;\in\; \mathbb{R} \quad : \quad x \;\neq\; 0 \quad \text{and} \quad x \;\neq\; 3 \,\}$$

The comma often written between the two conditions means **and**, to be precise.




The vertical dashed line at $x = 3$ marks where the composite blows up, and the small circle at the origin marks the point removed by Rule 1 — a value the simplified formula would happily accept, but the composition does not.

## 4 Exercise

**Exercise 8.1**

Given

$$f(x) = \frac{1}{x+1}, \qquad g(x) = \frac{1}{x},$$

find $(f \circ g)(x)$ and the domain of $f \circ g$.

Solve this quickly and check whether you have understood the concept of composition of functions and of the domain.

**Check your answers**

$$(f \circ g)(x) = \frac{1}{g(x)+1} = \frac{1}{\frac{1}{x}+1} = \frac{x}{1+x} \qquad \text{(assuming } x \neq 0\text{)}.$$

**Rule 1:** $g(x) = \dfrac{1}{x}$ needs $x \neq 0$, so $x = 0$ is excluded.

**Rule 2:** $f$ needs its argument $\neq -1$, so $g(x) = \dfrac{1}{x} \neq -1$, giving $x \neq -1$.

$$\text{Dom}(f \circ g) = \left\{ x \in \mathbb{R} \ : \ x \neq 0 \ \text{ and } \ x \neq -1 \right\}$$

## References and Acknowledgement

These notes are based on the video lectures of Mathematics–1 offered by the IIT Madras BS Degree Programme.

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